LAB 7: Fuzzing for Input Validation Bugs

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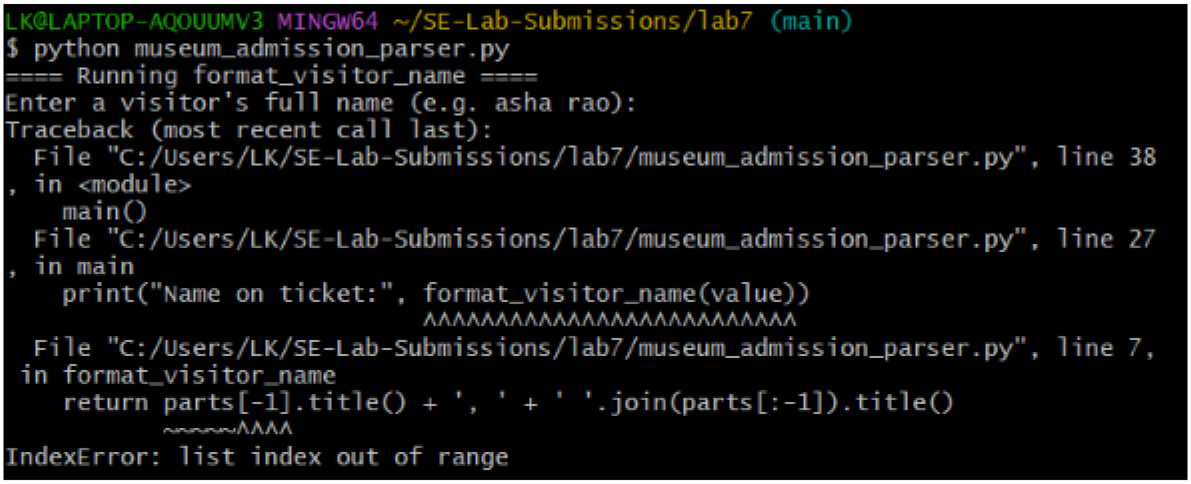
Class: 5J

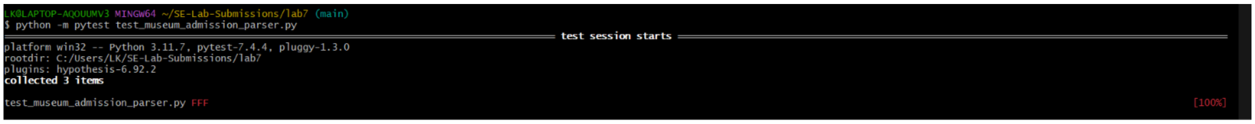
Github link: https://github.com/pranavyv/SE-Lab-Submissions/tree/main/lab7

Output of python museum_admission_parser.py on the buggy module

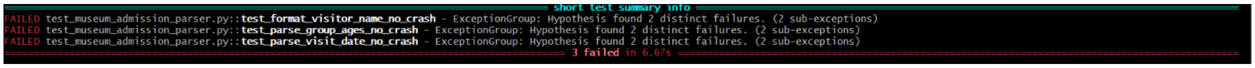


```
LK@LAPTOP-AQOUUMV3 MINGW64 ~/SE-Lab-Submissions/lab7 (main)
$ python museum_admission_parser.py
==== Running format_visitor_name ====
Enter a visitor's full name (e.g. asha rao):
Traceback (most recent call last):
  File "C:/Users/LK/SE-Lab-Submissions/lab7/museum_admission_parser.py", line 38
, in <module>
    main()
  File "C:/Users/LK/SE-Lab-Submissions/lab7/museum_admission_parser.py", line 27
, in main
    print("Name on ticket:", format_visitor_name(value))
                             ^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:/Users/LK/SE-Lab-Submissions/lab7/museum_admission_parser.py", line 7,
 in format_visitor_name
    return parts[-1].title() + ', ' + ' '.join(parts[:-1]).title()
           ~~~~~^^^^
IndexError: list index out of range
```

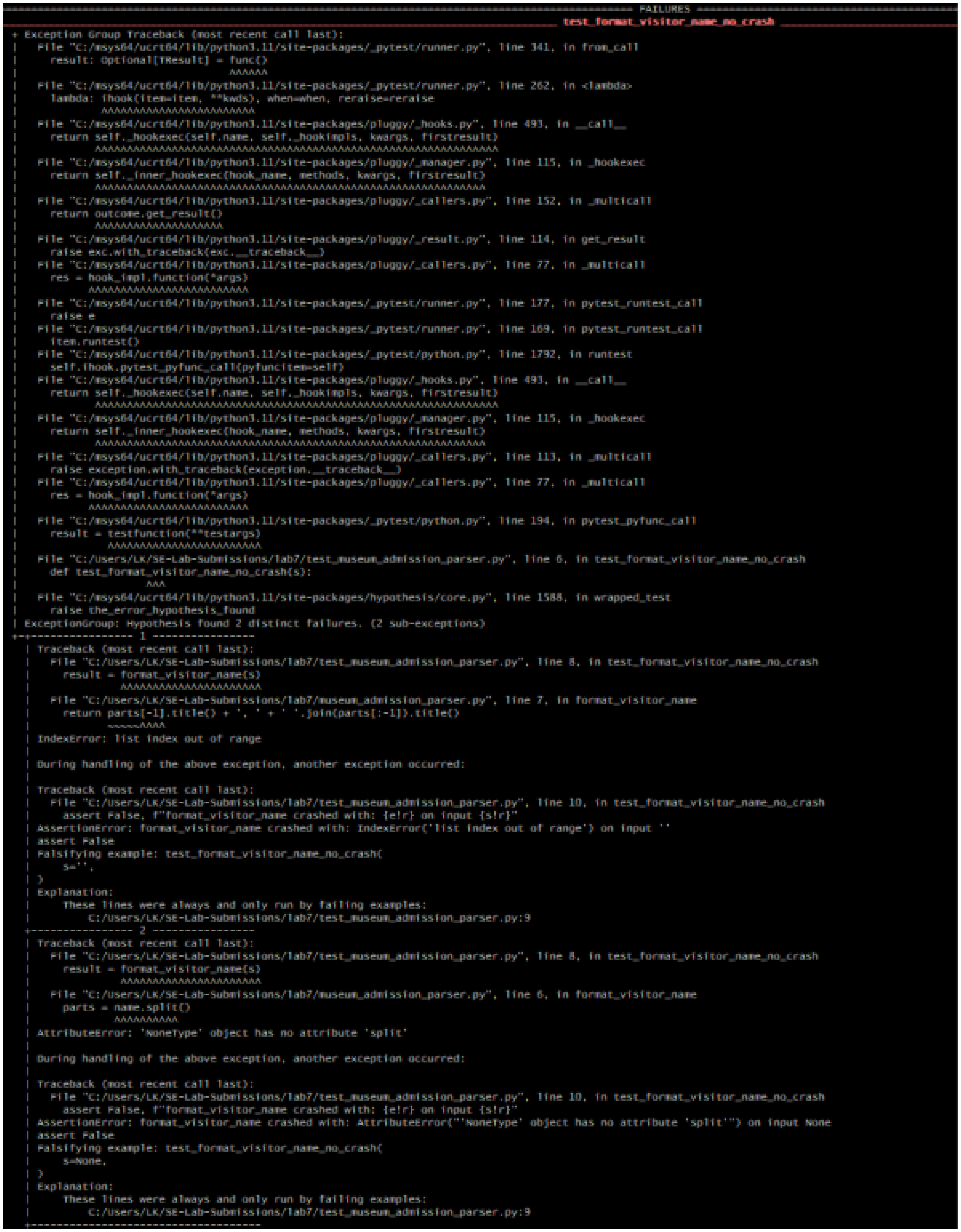
Output of pytest test_museum_admission_parser.py on the buggy module (tests failing)



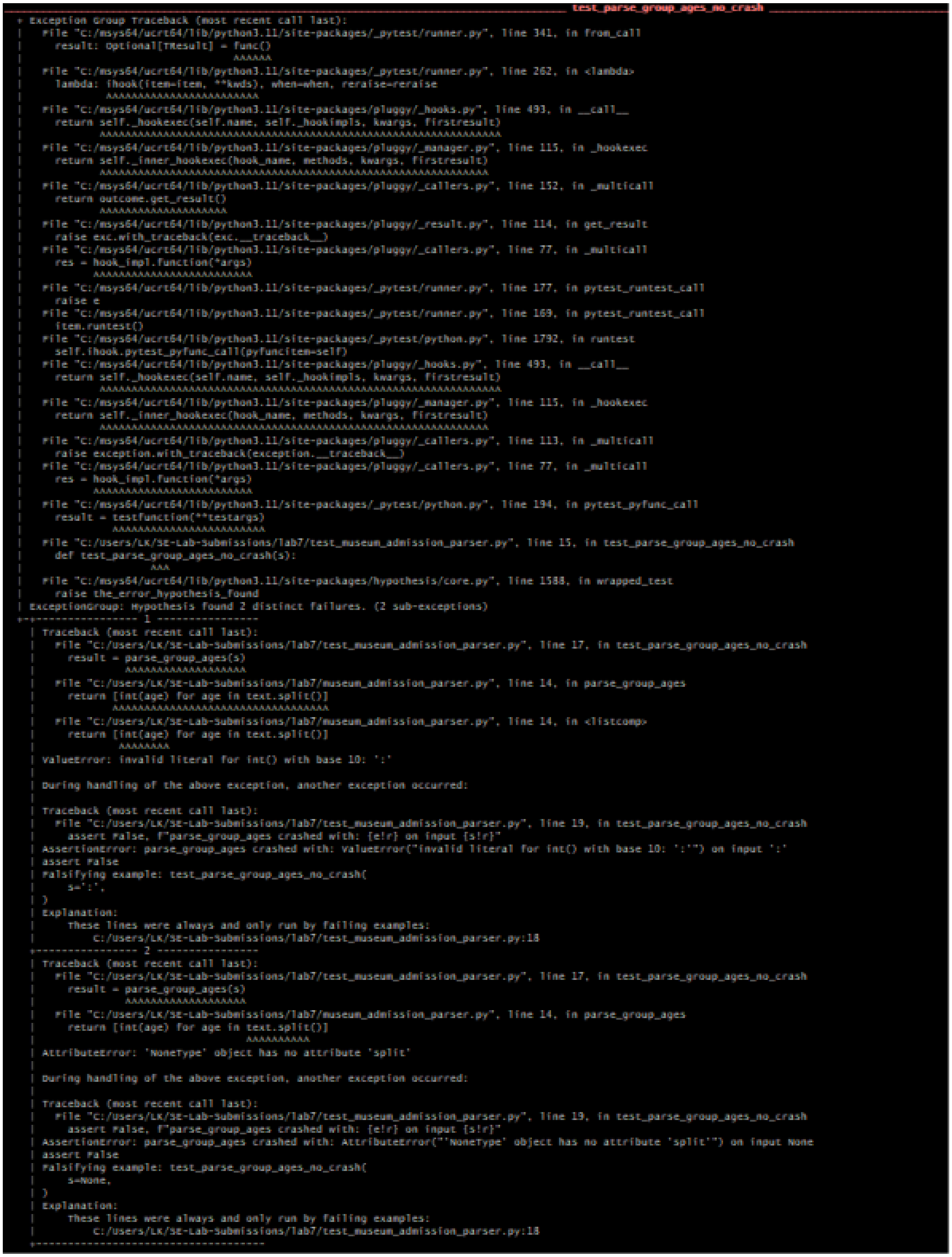
```
LK@LAPTOP-AQOUUMV3 MINGW64 ~/SE-Lab-Submissions/lab7 (main)
$ python -m pytest test_museum_admission_parser.py
============================================ test session starts ============================================
platform win32 -- Python 3.11.7, pytest-7.4.4, pluggy-1.3.0
rootdir: C:/Users/LK/SE-Lab-Submissions/lab7
plugins: hypothesis-6.92.2
collected 3 items

test_museum_admission_parser.py FFF                                                                    [100%]
```



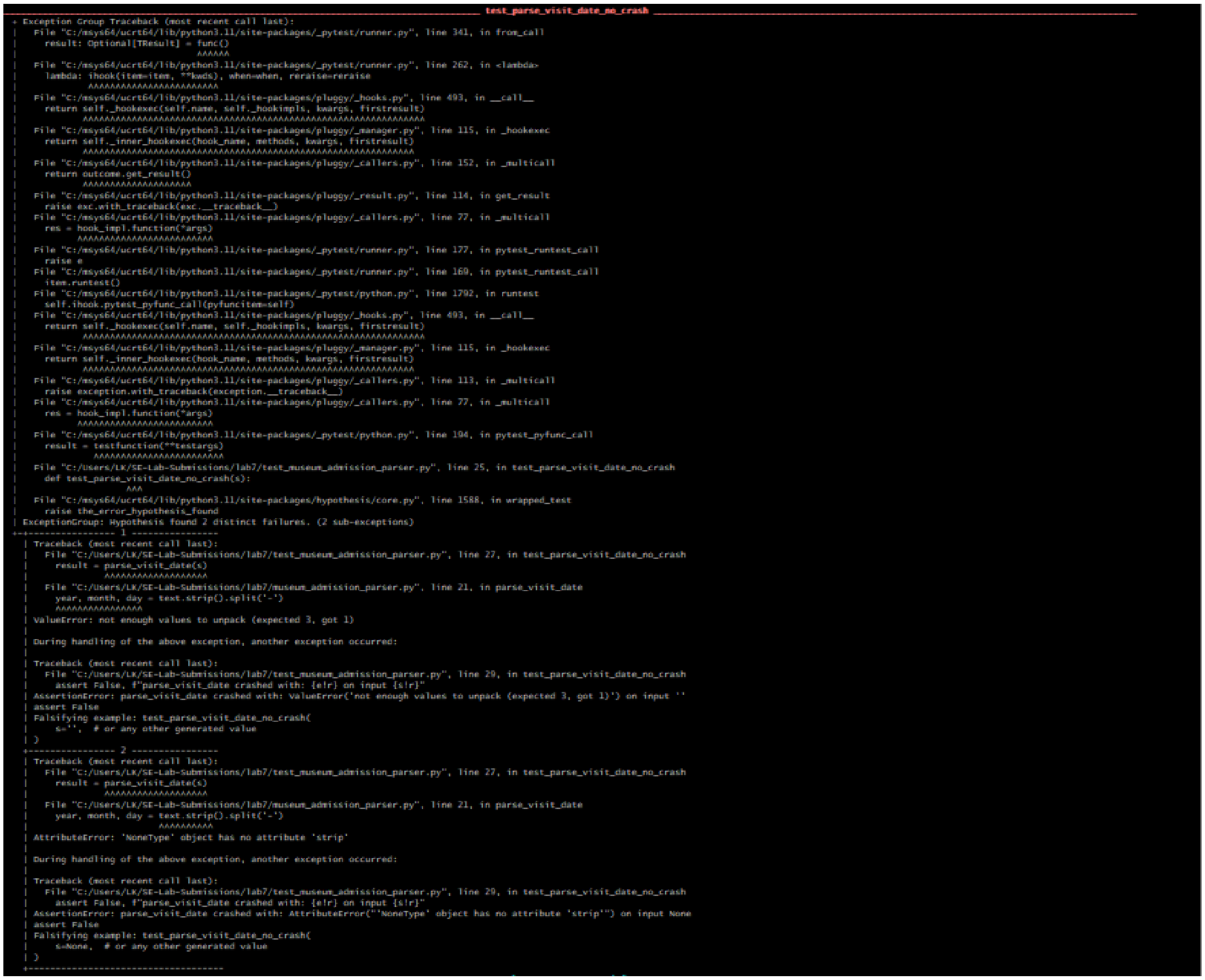
```
========================================== short test summary info ==========================================
FAILED test_museum_admission_parser.py::test_format_visitor_name_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_museum_admission_parser.py::test_parse_group_ages_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_museum_admission_parser.py::test_parse_visit_date_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
============================================== 3 failed in 6.67s ==============================================
```



```
================================================================================ FAILURES ================================================================================
_________________________________________________________________ test_format_visitor_name_no_crash _________________________________________________________________
+ Exception Group Traceback (most recent call last):
|   File "C:/msys64/ucrt64/lib/python3.11/site-packages/_pytest/runner.py", line 341, in from_call
|     result: Optional[TResult] = func()
|                                 ^^^^^^
|   File "C:/msys64/ucrt64/lib/python3.11/site-packages/_pytest/runner.py", line 262, in <lambda>
|     lambda: ihook(item=item, **kwds), when=when, reraise=reraise
|             ^^^^^^^^^^^^^^^^^^^^^^^^
|   File "C:/msys64/ucrt64/lib/python3.11/site-packages/pluggy/_hooks.py", line 493, in __call__
|     return self._hookexec(self.name, self._hookimpls, kwargs, firstresult)
|            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
|   File "C:/msys64/ucrt64/lib/python3.11/site-packages/pluggy/_manager.py", line 115, in _hookexec
|     return self._inner_hookexec(hook_name, methods, kwargs, firstresult)
|            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
|   File "C:/msys64/ucrt64/lib/python3.11/site-packages/pluggy/_callers.py", line 152, in _multicall
|     return outcome.get_result()
|            ^^^^^^^^^^^^^^^^^^^^
|   File "C:/msys64/ucrt64/lib/python3.11/site-packages/pluggy/_result.py", line 114, in get_result
|     raise exc.with_traceback(exc.__traceback__)
|   File "C:/msys64/ucrt64/lib/python3.11/site-packages/pluggy/_callers.py", line 77, in _multicall
|     res = hook_impl.function(*args)
|           ^^^^^^^^^^^^^^^^^^^^^^^^^
|   File "C:/msys64/ucrt64/lib/python3.11/site-packages/_pytest/runner.py", line 177, in pytest_runtest_call
|     raise e
|   File "C:/msys64/ucrt64/lib/python3.11/site-packages/_pytest/runner.py", line 169, in pytest_runtest_call
|     item.runtest()
|   File "C:/msys64/ucrt64/lib/python3.11/site-packages/_pytest/python.py", line 1792, in runtest
|     self.ihook.pytest_pyfunc_call(pyfuncitem=self)
|   File "C:/msys64/ucrt64/lib/python3.11/site-packages/pluggy/_hooks.py", line 493, in __call__
|     return self._hookexec(self.name, self._hookimpls, kwargs, firstresult)
|            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
|   File "C:/msys64/ucrt64/lib/python3.11/site-packages/pluggy/_manager.py", line 115, in _hookexec
|     return self._inner_hookexec(hook_name, methods, kwargs, firstresult)
|            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
|   File "C:/msys64/ucrt64/lib/python3.11/site-packages/pluggy/_callers.py", line 113, in _multicall
|     raise exception.with_traceback(exception.__traceback__)
|   File "C:/msys64/ucrt64/lib/python3.11/site-packages/pluggy/_callers.py", line 77, in _multicall
|     res = hook_impl.function(*args)
|           ^^^^^^^^^^^^^^^^^^^^^^^^^
|   File "C:/msys64/ucrt64/lib/python3.11/site-packages/_pytest/python.py", line 194, in pytest_pyfunc_call
|     result = testfunction(**testargs)
|              ^^^^^^^^^^^^^^^^^^^^^^^^
|   File "C:/Users/LK/SE-Lab-Submissions/lab7/test_museum_admission_parser.py", line 6, in test_format_visitor_name_no_crash
|     def test_format_visitor_name_no_crash(s):
|                    ^^^
|   File "C:/msys64/ucrt64/lib/python3.11/site-packages/hypothesis/core.py", line 1588, in wrapped_test
|     raise the_error_hypothesis_found
| ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
+-+---------------- 1 ----------------
  | Traceback (most recent call last):
  |   File "C:/Users/LK/SE-Lab-Submissions/lab7/test_museum_admission_parser.py", line 8, in test_format_visitor_name_no_crash
  |     result = format_visitor_name(s)
  |              ^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:/Users/LK/SE-Lab-Submissions/lab7/museum_admission_parser.py", line 7, in format_visitor_name
  |     return parts[-1].title() + ', ' + ' '.join(parts[:-1]).title()
  |            ~~~~~^^^^
  | IndexError: list index out of range
  |
  | During handling of the above exception, another exception occurred:
  |
  | Traceback (most recent call last):
  |   File "C:/Users/LK/SE-Lab-Submissions/lab7/test_museum_admission_parser.py", line 10, in test_format_visitor_name_no_crash
  |     assert False, f"format_visitor_name crashed with: {e!r} on input {s!r}"
  | AssertionError: format_visitor_name crashed with: IndexError('list index out of range') on input ''
  | assert False
  | Falsifying example: test_format_visitor_name_no_crash(
  |     s='',
  | )
  | Explanation:
  |     These lines were always and only run by failing examples:
  |         C:/Users/LK/SE-Lab-Submissions/lab7/test_museum_admission_parser.py:9
  +---------------- 2 ----------------
  | Traceback (most recent call last):
  |   File "C:/Users/LK/SE-Lab-Submissions/lab7/test_museum_admission_parser.py", line 8, in test_format_visitor_name_no_crash
  |     result = format_visitor_name(s)
  |              ^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:/Users/LK/SE-Lab-Submissions/lab7/museum_admission_parser.py", line 6, in format_visitor_name
  |     parts = name.split()
  |             ^^^^^^^^^^
  | AttributeError: 'NoneType' object has no attribute 'split'
  |
  | During handling of the above exception, another exception occurred:
  |
  | Traceback (most recent call last):
  |   File "C:/Users/LK/SE-Lab-Submissions/lab7/test_museum_admission_parser.py", line 10, in test_format_visitor_name_no_crash
  |     assert False, f"format_visitor_name crashed with: {e!r} on input {s!r}"
  | AssertionError: format_visitor_name crashed with: AttributeError("'NoneType' object has no attribute 'split'") on input None
  | assert False
  | Falsifying example: test_format_visitor_name_no_crash(
  |     s=None,
  | )
  | Explanation:
  |     These lines were always and only run by failing examples:
  |         C:/Users/LK/SE-Lab-Submissions/lab7/test_museum_admission_parser.py:9
  +------------------------------------
```

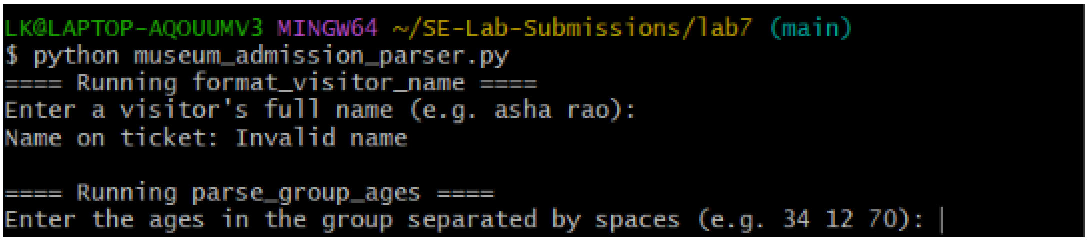


```
________________________________________________________________________________________ test_parse_group_ages_no_crash ________________________________________________________________________________________
  + Exception Group Traceback (most recent call last):
  |   File "C:/msys64/ucrt64/lib/python3.11/site-packages/_pytest/runner.py", line 341, in from_call
  |     result: Optional[TResult] = func()
  |                                 ^^^^^^
  |   File "C:/msys64/ucrt64/lib/python3.11/site-packages/_pytest/runner.py", line 262, in <lambda>
  |     lambda: ihook(item=item, **kwds), when=when, reraise=reraise
  |             ^^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:/msys64/ucrt64/lib/python3.11/site-packages/pluggy/_hooks.py", line 493, in __call__
  |     return self._hookexec(self.name, self._hookimpls, kwargs, firstresult)
  |            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:/msys64/ucrt64/lib/python3.11/site-packages/pluggy/_manager.py", line 115, in _hookexec
  |     return self._inner_hookexec(hook_name, methods, kwargs, firstresult)
  |            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:/msys64/ucrt64/lib/python3.11/site-packages/pluggy/_callers.py", line 152, in _multicall
  |     return outcome.get_result()
  |            ^^^^^^^^^^^^^^^^^^^^
  |   File "C:/msys64/ucrt64/lib/python3.11/site-packages/pluggy/_result.py", line 114, in get_result
  |     raise exc.with_traceback(exc.__traceback__)
  |   File "C:/msys64/ucrt64/lib/python3.11/site-packages/pluggy/_callers.py", line 77, in _multicall
  |     res = hook_impl.function(*args)
  |           ^^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:/msys64/ucrt64/lib/python3.11/site-packages/_pytest/runner.py", line 177, in pytest_runtest_call
  |     raise e
  |   File "C:/msys64/ucrt64/lib/python3.11/site-packages/_pytest/runner.py", line 169, in pytest_runtest_call
  |     item.runtest()
  |   File "C:/msys64/ucrt64/lib/python3.11/site-packages/_pytest/python.py", line 1792, in runtest
  |     self.ihook.pytest_pyfunc_call(pyfuncitem=self)
  |   File "C:/msys64/ucrt64/lib/python3.11/site-packages/pluggy/_hooks.py", line 493, in __call__
  |     return self._hookexec(self.name, self._hookimpls, kwargs, firstresult)
  |            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:/msys64/ucrt64/lib/python3.11/site-packages/pluggy/_manager.py", line 115, in _hookexec
  |     return self._inner_hookexec(hook_name, methods, kwargs, firstresult)
  |            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:/msys64/ucrt64/lib/python3.11/site-packages/pluggy/_callers.py", line 113, in _multicall
  |     raise exception.with_traceback(exception.__traceback__)
  |   File "C:/msys64/ucrt64/lib/python3.11/site-packages/pluggy/_callers.py", line 77, in _multicall
  |     res = hook_impl.function(*args)
  |           ^^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:/msys64/ucrt64/lib/python3.11/site-packages/_pytest/python.py", line 194, in pytest_pyfunc_call
  |     result = testfunction(**testargs)
  |              ^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:/Users/LK/SE-Lab-Submissions/lab7/test_museum_admission_parser.py", line 15, in test_parse_group_ages_no_crash
  |     def test_parse_group_ages_no_crash(s):
  |                    ^^^
  |   File "C:/msys64/ucrt64/lib/python3.11/site-packages/hypothesis/core.py", line 1588, in wrapped_test
  |     raise the_error_hypothesis_found
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:/Users/LK/SE-Lab-Submissions/lab7/test_museum_admission_parser.py", line 17, in test_parse_group_ages_no_crash
    |     result = parse_group_ages(s)
    |              ^^^^^^^^^^^^^^^^^^^
    |   File "C:/Users/LK/SE-Lab-Submissions/lab7/museum_admission_parser.py", line 14, in parse_group_ages
    |     return [int(age) for age in text.split()]
    |            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
    |   File "C:/Users/LK/SE-Lab-Submissions/lab7/museum_admission_parser.py", line 14, in <listcomp>
    |     return [int(age) for age in text.split()]
    |             ^^^^^^^^
    | ValueError: invalid literal for int() with base 10: ':'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:/Users/LK/SE-Lab-Submissions/lab7/test_museum_admission_parser.py", line 19, in test_parse_group_ages_no_crash
    |     assert False, f"parse_group_ages crashed with: {e!r} on input {s!r}"
    | AssertionError: parse_group_ages crashed with: ValueError("invalid literal for int() with base 10: ':'") on input ':'
    | assert False
    | Falsifying example: test_parse_group_ages_no_crash(
    |     s=':',
    | )
    | Explanation:
    |     These lines were always and only run by failing examples:
    |         C:/Users/LK/SE-Lab-Submissions/lab7/test_museum_admission_parser.py:18
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:/Users/LK/SE-Lab-Submissions/lab7/test_museum_admission_parser.py", line 17, in test_parse_group_ages_no_crash
    |     result = parse_group_ages(s)
    |              ^^^^^^^^^^^^^^^^^^^
    |   File "C:/Users/LK/SE-Lab-Submissions/lab7/museum_admission_parser.py", line 14, in parse_group_ages
    |     return [int(age) for age in text.split()]
    |                                 ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:/Users/LK/SE-Lab-Submissions/lab7/test_museum_admission_parser.py", line 19, in test_parse_group_ages_no_crash
    |     assert False, f"parse_group_ages crashed with: {e!r} on input {s!r}"
    | AssertionError: parse_group_ages crashed with: AttributeError("'NoneType' object has no attribute 'split'") on input None
    | assert False
    | Falsifying example: test_parse_group_ages_no_crash(
    |     s=None,
    | )
    | Explanation:
    |     These lines were always and only run by failing examples:
    |         C:/Users/LK/SE-Lab-Submissions/lab7/test_museum_admission_parser.py:18
    +------------------------------------
```



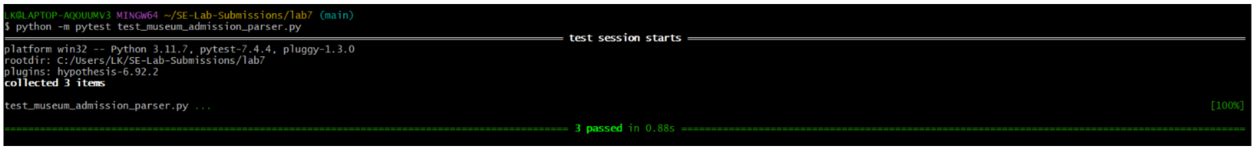
```
____________________________________________________________________________________ test_parse_visit_date_no_crash _____________________________________________________________________________________
+ Exception Group Traceback (most recent call last):
|   File "C:/msys64/ucrt64/lib/python3.11/site-packages/_pytest/runner.py", line 341, in from_call
|     result: Optional[TResult] = func()
|                                 ^^^^^^
|   File "C:/msys64/ucrt64/lib/python3.11/site-packages/_pytest/runner.py", line 262, in <lambda>
|     lambda: ihook(item=item, **kwds), when=when, reraise=reraise
|             ^^^^^^^^^^^^^^^^^^^^^^^^
|   File "C:/msys64/ucrt64/lib/python3.11/site-packages/pluggy/_hooks.py", line 493, in __call__
|     return self._hookexec(self.name, self._hookimpls, kwargs, firstresult)
|            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
|   File "C:/msys64/ucrt64/lib/python3.11/site-packages/pluggy/_manager.py", line 115, in _hookexec
|     return self._inner_hookexec(hook_name, methods, kwargs, firstresult)
|            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
|   File "C:/msys64/ucrt64/lib/python3.11/site-packages/pluggy/_callers.py", line 152, in _multicall
|     return outcome.get_result()
|            ^^^^^^^^^^^^^^^^^^^^
|   File "C:/msys64/ucrt64/lib/python3.11/site-packages/pluggy/_result.py", line 114, in get_result
|     raise exc.with_traceback(exc.__traceback__)
|   File "C:/msys64/ucrt64/lib/python3.11/site-packages/pluggy/_callers.py", line 77, in _multicall
|     res = hook_impl.function(*args)
|           ^^^^^^^^^^^^^^^^^^^^^^^^^
|   File "C:/msys64/ucrt64/lib/python3.11/site-packages/_pytest/runner.py", line 177, in pytest_runtest_call
|     raise e
|   File "C:/msys64/ucrt64/lib/python3.11/site-packages/_pytest/runner.py", line 169, in pytest_runtest_call
|     item.runtest()
|   File "C:/msys64/ucrt64/lib/python3.11/site-packages/_pytest/python.py", line 1792, in runtest
|     self.ihook.pytest_pyfunc_call(pyfuncitem=self)
|   File "C:/msys64/ucrt64/lib/python3.11/site-packages/pluggy/_hooks.py", line 493, in __call__
|     return self._hookexec(self.name, self._hookimpls, kwargs, firstresult)
|            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
|   File "C:/msys64/ucrt64/lib/python3.11/site-packages/pluggy/_manager.py", line 115, in _hookexec
|     return self._inner_hookexec(hook_name, methods, kwargs, firstresult)
|            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
|   File "C:/msys64/ucrt64/lib/python3.11/site-packages/pluggy/_callers.py", line 113, in _multicall
|     raise exception.with_traceback(exception.__traceback__)
|   File "C:/msys64/ucrt64/lib/python3.11/site-packages/pluggy/_callers.py", line 77, in _multicall
|     res = hook_impl.function(*args)
|           ^^^^^^^^^^^^^^^^^^^^^^^^^
|   File "C:/msys64/ucrt64/lib/python3.11/site-packages/_pytest/python.py", line 194, in pytest_pyfunc_call
|     result = testfunction(**testargs)
|              ^^^^^^^^^^^^^^^^^^^^^^^^
|   File "C:/Users/LK/SE-Lab-Submissions/lab7/test_museum_admission_parser.py", line 25, in test_parse_visit_date_no_crash
|     def test_parse_visit_date_no_crash(s):
|                    ^^^
|   File "C:/msys64/ucrt64/lib/python3.11/site-packages/hypothesis/core.py", line 1588, in wrapped_test
|     raise the_error_hypothesis_found
| ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
+-+---------------- 1 ----------------
  | Traceback (most recent call last):
  |   File "C:/Users/LK/SE-Lab-Submissions/lab7/test_museum_admission_parser.py", line 27, in test_parse_visit_date_no_crash
  |     result = parse_visit_date(s)
  |              ^^^^^^^^^^^^^^^^^^^
  |   File "C:/Users/LK/SE-Lab-Submissions/lab7/museum_admission_parser.py", line 21, in parse_visit_date
  |     year, month, day = text.strip().split('-')
  |     ^^^^^^^^^^^^^^^^
  | ValueError: not enough values to unpack (expected 3, got 1)
  |
  | During handling of the above exception, another exception occurred:
  |
  | Traceback (most recent call last):
  |   File "C:/Users/LK/SE-Lab-Submissions/lab7/test_museum_admission_parser.py", line 29, in test_parse_visit_date_no_crash
  |     assert False, f"parse_visit_date crashed with: {e!r} on input {s!r}"
  | AssertionError: parse_visit_date crashed with: ValueError('not enough values to unpack (expected 3, got 1)') on input ''
  | assert False
  | Falsifying example: test_parse_visit_date_no_crash(
  |     s='',  # or any other generated value
  | )
  +---------------- 2 ----------------
  | Traceback (most recent call last):
  |   File "C:/Users/LK/SE-Lab-Submissions/lab7/test_museum_admission_parser.py", line 27, in test_parse_visit_date_no_crash
  |     result = parse_visit_date(s)
  |              ^^^^^^^^^^^^^^^^^^^
  |   File "C:/Users/LK/SE-Lab-Submissions/lab7/museum_admission_parser.py", line 21, in parse_visit_date
  |     year, month, day = text.strip().split('-')
  |                        ^^^^^^^^^^
  | AttributeError: 'NoneType' object has no attribute 'strip'
  |
  | During handling of the above exception, another exception occurred:
  |
  | Traceback (most recent call last):
  |   File "C:/Users/LK/SE-Lab-Submissions/lab7/test_museum_admission_parser.py", line 29, in test_parse_visit_date_no_crash
  |     assert False, f"parse_visit_date crashed with: {e!r} on input {s!r}"
  | AssertionError: parse_visit_date crashed with: AttributeError("'NoneType' object has no attribute 'strip'") on input None
  | assert False
  | Falsifying example: test_parse_visit_date_no_crash(
  |     s=None,  # or any other generated value
  | )
  +------------------------------------
```

Output of python museum_admission_parser.py on the fixed module



```
LK@LAPTOP-AQOUUMV3 MINGW64 ~/SE-Lab-Submissions/lab7 (main)
$ python museum_admission_parser.py
==== Running format_visitor_name ====
Enter a visitor's full name (e.g. asha rao):
Name on ticket: Invalid name

==== Running parse_group_ages ====
Enter the ages in the group separated by spaces (e.g. 34 12 70): |
```

Output of pytest test_museum_admission_parser.py on the fixed module (all tests passing)



```
LK@LAPTOP-AQOUUMV3 MINGW64 ~/SE-Lab-Submissions/lab7 (main)
$ python -m pytest test_museum_admission_parser.py
================================================ test session starts ================================================
platform win32 -- Python 3.11.7, pytest-7.4.4, pluggy-1.3.0
rootdir: C:/Users/LK/SE-Lab-Submissions/lab7
plugins: hypothesis-6.92.2
collected 3 items

test_museum_admission_parser.py ...                                                                            [100%]

================================================= 3 passed in 0.88s =================================================
```